Actor Checkpoint Restoration in Asterix PPO: A Reference-Aware Study of Recovery Cost

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Abstract

A temporarily impaired reinforcement-learning agent can lag a clean control because of foregone progress, additional recovery cost, or both. We distinguish same-update gap attribution from interventions on detected harm relative to a historical pause benchmark. Our primary evidence is one preregistered Asterix PPO study with 32 new training seeds, reward-sign corruption and a fixed 200-update recovery horizon. Native pause-relative recovery cost is $-1.72\,[-1.94, -1.50]$ return points. Restoring only the pre-impairment actor, without the clean fork's additional progress, improves recovery by $+1.58\,[+1.37, +1.79]$. A single four-component practical-effect criterion, requiring native and late harm, rescue and reciprocal transfer, passes at $p = 3.32 \times 10^{-8}$, $\alpha = .01$. The intervention removes 91.8% of the measured mean cost (paired-bootstrap 95% interval 86.3–97.5%), not 91.8% of naturally stored damage. A small residual remains; matched-state practical equivalence fails. Specified optimizer, world-reset and second-evaluation checks retain positive rescue. We provide an explicit evidence hierarchy for the heterogeneous earlier studies, without paper-wide multiplicity control. The conclusion is a setting- and intervention-specific repair result, not a general RL recovery law, proof of irreversible damage, a new-task plasticity result or a unique internal localization of impairment.

1 Introduction

A clean agent continues learning while an impaired copy falls behind. Replacing the impaired copy's parameters with the clean copy's later parameters may close this gap simply by importing the progress it missed. Such a transplant can identify a useful intervention without identifying a cause of recovery cost *beyond* lost training time. A reference trajectory is therefore part of the scientific claim, not a cosmetic plotting choice.

We ask a narrower question: *in one selected setting with detected pause-relative harm, does the learner's own pre-impairment actor checkpoint repair its subsequent recovery cost?* In Asterix PPO, a fixed 32-seed confirmation supports this intervention. The actor donor predates reward corruption; all other primary removal-time fields come from the naturally impaired learner. The primary requires four effects together, rather than selecting the most favourable endpoint. It is met, with native cost negative and actor rescue positive in all 32 seeds. The result is substantial but not complete restoration, and it does not locate a unique internal representation of impairment.

The main text presents the reference framework, this one study, its scope and the evidence hierarchy. The accompanying Supplementary Material preserves the full historical protocols, all result tables, negative/inconclusive primaries, context checks, registration timelines and deviations. Earlier experiments motivate the selected case; they are not pooled into its primary or treated as equally strong independent confirmations.

Relation to prior work. Critical-period work motivates studying effects of temporary deficits, but does not establish irreversibility in this assay (Achille et al., 2019; Kleinman et al., 2024). Parameter resets, distillation and plasticity studies address related recovery and learning-capacity questions (Igl et al., 2021; Nikishin et al., 2022; Lyle et al., 2023; Dohare et al., 2024). Optimizer state can also matter in other settings

(Dohare et al., 2023; Xu, 2026). Our contribution is not that transplantation or actor checkpoints are new in general. It is the explicit separation of reference-dependent targets and a preregistered, naturally continued, past-actor intervention in a case with detected beyond-time recovery cost. Preserved capacity on new tasks is a different question, not measured here.

2 A reference-aware framework

2.1 Compact glossary

Table 1: Terminology used throughout. Each effect is conditional on its reference, future context, evaluation protocol and recovery horizon.

term	operational meaning
Tier I	Component-state intervention on the gap to a clean agent at the *same training update*. This gap includes foregone clean-run progress.
Tier II	A direct pause/onset-reference intervention coupled to detected *native pause-relative harm*. It supports an intervention-specific statement about that harm, not unique natural storage.
Pause reference	Here, the historical clean trace shifted back by the impairment duration: $H_i(k) = C_i(w_0+k)$. A matched onset-state learner is a separate intervention reference, not assumed identical to H.
Native recovery cost	The recovery AUC difference between the natural impaired continuation D_i and H_i. Negative means worse evaluation return. "Native" excludes an artificial clean-world continuation substituted for D.
State surgery	A specified field-level replacement of learner state, followed by ordinary learning. The causal effect includes downstream changes in actions, experience, critic and optimizer.
Shapley allocation	An allocation of a measured factorial gap: average each component's marginal effect over all insertion orders. It splits interactions by an ordering rule and sums to the selected gap; it is not an information decomposition.

2.2 What the reported AUC measures

Let w_0 be impairment onset, $w_1 = w_0 + L$ removal, and $V_i^X(k)$ the mean unrestricted evaluation return of seed i, arm X, at recovery checkpoint k. In the primary study,

$$\mathcal{K} = \{10, 20, \ldots, 200\}, \qquad A_i(X) = \frac{1}{20} \sum_{k \in \mathcal{K}} V_i^X(k). \tag{1}$$

Our "recovery AUC" is this **discrete time-average**, not an unnormalised integral or a trapezoidal area. It retains units of **evaluation episode-return points**. We average paired seed differences, not episodes or checkpoints as independent observations. The removal evaluation $k = 0$ is plotted but excluded from AUC. The late endpoint averages the last five checkpoints, $k = 160, 170, 180, 190, 200$.

The native cost and same-update gap use different benchmarks:

$$E_i^{\text{native}} = A_i(D) - A_i(H), \qquad H_i(k) = C_i(w_0 + k), \tag{2}$$

$$E_i^{\text{same}} = A_i(D) - A_i(C_{\text{same}}), \qquad C_{\text{same},i}(k) = C_i(w_1 + k). \tag{3}$$

A negative E^{native} is extra cost relative to this historical lost-time benchmark. It is not a guarantee that a literal paused runner in every future world would follow H. The matched onset-state reference is measured separately for precisely this reason.

Scale and practical meaning. An Asterix effect of -1.72 means that average evaluation return is 1.72 points below the specified reference across those 20 checkpoints. It is *not* 1.72 training steps, the total

training reward lost, or a time-to-recovery estimate. One PPO update is 8,192 environment steps, so the 200-update recovery spans 1,638,400 steps. The registered .50 AUC margin was calibrated against a prior clean score of about 6.0 (roughly 8%); the .30 late margin was 5% of that same prior scale. These calibrations are specific to this game. Raw AUC magnitudes from SpaceInvaders, CartPole or PQN studies cannot be ranked or pooled as if they shared reward scale, evaluation policy and recovery grid. Cross-algorithm recovery comparisons additionally require matching *environment steps*, not update counts.

2.3 Intervention effects are not storage fractions

A clean-fork swap, a past-actor swap, a factorial main effect and a Shapley allocation answer different questions. The primary study uses a *background-specific* actor swap in the natural impaired context; it is not a full three-component allocation. Its repaired fraction is

$$\rho = \frac{\overline{A(R) - A(D)}}{-\overline{E^{\text{native}}}}, \tag{4}$$

where R denotes actor restoration. It is a ratio of mean intervention effects, not the average of seedwise percentages, a Shapley value, or the proportion of impairment encoded in actor weights. Changed future experience is part of the treatment effect, not held fixed by definition.

3 The preregistered Asterix study

Fixed selected condition. PPO (Schulman et al., 2017) uses separate 2×128 ReLU actor and critic MLPs, Adam at constant learning rate 5×10^{-4}, 64 environments × 128 steps, four epochs and eight minibatches. The environment is Asterix-MinAtar in gymnax (Young & Tian, 2019; Lange, 2022), implemented in JAX in the PureJaxRL style (Lu et al., 2022). Training rewards have their signs flipped during updates $[390, 440)$ (409,600 steps), followed by 200 unrestricted updates. Evaluation always uses true reward, clean observations and the full action set. Every ten updates, 64 stochastic-policy evaluation episodes use key 10000. The analysis pairs independent training seeds conditional on this fixed evaluation batch.

All 32 unused IDs 1200–1231 and all 11 requested cells were registered before target data. A previous independent eight-seed Asterix cohort motivated this selected condition; it is not pooled into the primary and does not make this a new-task generalization test. No low-competence seed was removed, no primary analysis occurred before all 32 completed, and no setting, endpoint, sample-size or evaluation-key switch was used.

Donors and natural continuation. P is the shared onset state at update 390, D the impaired state at 440, and C the unimpaired fork at 440. The main repair replaces *only actor parameters in D with those from P*; it does not use the extra-progress actor from C. The removal-time critic, Adam state, environment, observations, future RNG and counters remain those of D. The all-impaired cell is asserted equal over the complete runner-state tree to untouched natural D. Actor/critic/optimizer composition is explicit in code; after intervention these fields and collected experience can diverge normally.

For reciprocal transfer, the impaired actor is inserted into a learner with onset critic parameters and Adam state, under the same natural D world/RNG/counter context. We denote that matched learner reference by P_D: it is *not* the full onset runner P or the historical trajectory H. In archived pause-cell codes, C means an onset donor and D an impaired donor: DDD is native, CDD actor-restored, CCC the matched reference, and DCC reciprocal impaired-actor transfer.

One primary claim, four required parts. The registration fixes four one-sided paired-t tests against practical-effect thresholds (Table 2). The single intersection–union claim uses $p_{\text{claim}} = \max_j p_j$ at $\alpha = .01$, with an additional requirement of at least 24/32 native-negative and 24/32 rescue-positive seeds. All parts must pass. Intervals are nominal training-seed intervals, with 99% intervals, exact signs, paired bootstrap and simultaneous-bound sensitivities retained in the supplement and analysis artifact (Henderson et al., 2018; Agarwal et al., 2021). The fixed second evaluation key 54321, common-fresh-Adam factorial and reset-world rescue are prespecified *secondary* checks; they cannot replace a failed primary.

Table 2: The single registered primary, $n = 32$. All four threshold-shifted tests and the consistency gate are required; $p_{\text{claim}} = 3.319 \times 10^{-8}$. Units are average evaluation-return points (Section 2.2). Brackets are nominal 95% paired-seed t-intervals.

contrast	mean [95% interval]	required mean	shifted one-sided p
Native AUC, $D - H$	$-1.722\,[-1.940, -1.504]$	$< -.50$	6.02×10^{-13}
Late native cost, $D - H$	$-1.187\,[-1.444, -.930]$	$< -.30$	3.32×10^{-8}
Actor-only rescue, $R - D$	$+1.580\,[+1.371, +1.790]$	$> +.50$	4.79×10^{-12}
Reciprocal transfer vs. P_D	$-1.636\,[-1.828, -1.445]$	$< -.50$	1.39×10^{-13}

Table 3: Prespecified secondary checks, $n = 32$, nominal 95% paired-seed intervals. TOST uses the same $\pm.30$ practical margin for both repair residuals; its outcome depends on the reference. These checks are not a new multiplicity-controlled primary family.

contrast	mean [95% interval]	interpretation
Fresh-Adam actor rescue	$+1.626\,[+1.420, +1.832]$	positive rescue interval
Fresh-Adam critic main effect	$-.129\,[-.229, -.029]$	smaller, not absent
Reset-world actor rescue	$+1.515\,[+1.308, +1.722]$	specified alternative context
Second-key actor rescue	$+1.398\,[+1.227, +1.569]$	key 54321; not pooled
Restored minus historical H	$-.141\,[-.246, -.037]$	TOST $p = .00211$: margin met
Restored minus matched P_D	$-.231\,[-.370, -.091]$	TOST $p = .160$: not met
Matched P_D minus H	$+.089\,[-.035, +.213]$	practical agreement, not identity

4 Results: most cost repaired, not full restoration

The compound criterion is met. Native AUC is negative in 32/32 seeds and actor rescue positive in 32/32, exceeding the registered 24/32 requirements. The late mean clears its practical margin; 31/32 late contrasts are negative, not every seed. Figure 1 shows every primary seed effect, both reference trajectories and the fixed robustness checks.

The past-actor intervention removes **91.8% of the mean native historical-pause cost**, with paired-bootstrap 95% interval 86.3–97.5%. This number describes the ratio ρ in this one experiment. It is neither "91.8% of damage stored in the actor" nor a universally applicable restoration rate.

Robustness does not imply neutrality or full restoration. All secondary results remain visible in Table 3. Fresh Adam, a reset world and the second evaluation key retain positive rescue intervals. The smaller negative fresh-Adam critic effect is not declared absent. Neither a nonsignificant context difference nor positive rescue in another context establishes context neutrality. The second key is not selected or pooled into the primary.

The repaired-minus-historical residual is small but negative. Its secondary $\pm.30$ TOST margin passes, while the same margin against P_D *fails* (Lakens, 2017). Practical-margin agreement is not literal equality. In particular, positive rescue and a 91.8% reduction do not establish complete recovery, even though the primary compound passes.

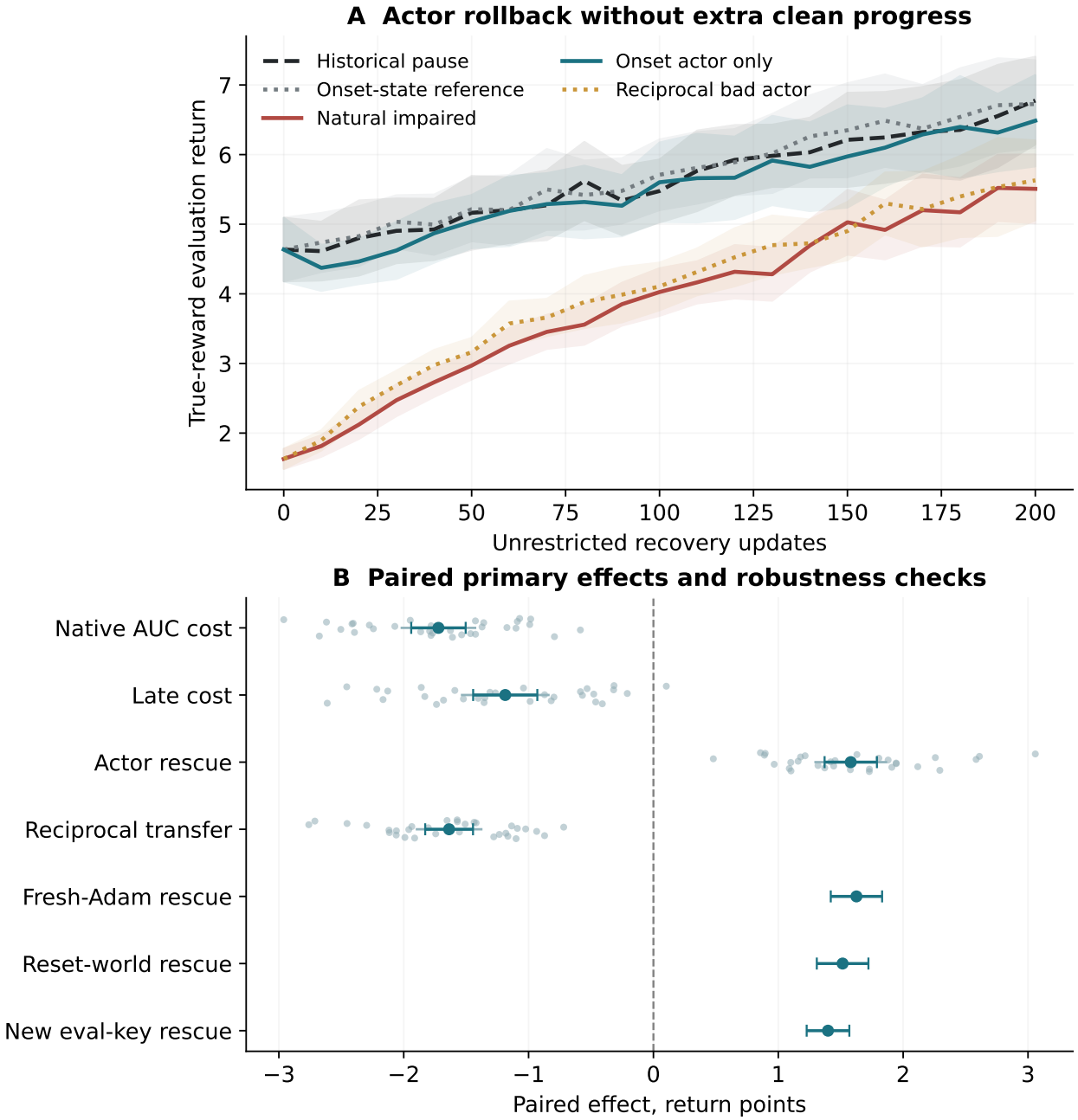


Figure 1: **One selected-condition actor-restoration result.** A: evaluations use true reward; the historical pause trace, matched onset-state reference, native learner, onset-actor repair and reciprocal actor transfer are distinct arms. The plotted removal point is excluded from AUC. Bands are nominal pointwise 99% seed intervals, not simultaneous trajectory bands. B: all 32 paired primary effects (small points), means and 95% intervals, with thin 99% primary whiskers. Secondary rescues are shown without promoting them to additional primary claims. Worlds and non-actor fields match *at removal* for the primary repair; downstream trajectories may diverge.

5 Evidence hierarchy across the research record

The primary result above is the strongest evidence for the *named intervention*. The broader record is heterogeneous in sample size, design status, reference and inferential target. Table 4 prevents a favourable result in one tier from upgrading an unresolved result in another. A preregistered design is not automatically a successful outcome. New contrasts on reused histories are not independent-seed replications, and selected eligible conditions are not a task-prevalence sample.

Table 4: Evidence hierarchy. Status attaches to the listed estimand and study, not to a universal carrier mechanism. Full values, references, N, families and deviations remain in the supplement and machine-readable inference map.

status	evidence and boundary
Strongest registered support	The present 32-seed Asterix PPO compound meets its fixed $\alpha = .01$ rule: native and late pause-relative harm, past-actor rescue and reciprocal transfer. One setting and one horizon.
Context-specific support	Prior independent Asterix 8 (Tier II) and actuator-noise 8 (Tier I) meet their fixed five-claim Phase-13 Holm criteria. The latter does *not* show native pause-relative harm. Both are selected conditions, not a task-prevalence sample.
Exploratory / descriptive	Early timing and four-seed reset/drift studies are exploratory. PPO's pooled gap allocation ($n = 12$) includes four pilot and eight new registered IDs. PQN-R network-block tests ($n = 12/8$, parameters plus RAdam) remain exploratory; Freeway has only 4 complete of 8 planned. Reused histories are not fresh-seed replications.
Narrow practical bound	Matched replay identity, $n = 32$: selected ± 1 TOST $p = .0495$; $\pm .75$ fails. Smaller effects remain possible. This is not absence of a data effect.
Unresolved primaries / mechanisms	Independent CartPole 8's stronger compound fails Phase-13 correction ($p = .0931$). The controlled-label assay 8 meets neither its substantial-harm nor equivalence rule. A buffer-through-learning mechanism remains interpretive; no secondary contrast rescues these primaries.
Prospective utility not validated	Held-out 16 frozen acute+KL prediction has MSE 68.78 versus 32.27 for acute-only and fails its utility criterion. Earlier KL associations do not establish useful future-failure prediction or a causal mechanism.

No paper-wide error-control claim. The 32-seed compound is one separate registered $\alpha = .01$ claim. Only the five Phase-13 primaries share their specified Holm family (Holm, 1979); older studies have other local families, exploratory analyses and disclosed interim looks. We neither combine these into a retrospective paper-wide false-positive guarantee nor count favourable panels as exchangeable independent confirmations. The main SpaceInvaders movement result, for example, has historical-pause AUC $+1.9\,[-.7, +4.4]$, $n = 12$: harm worse than -3 is excluded, but equivalence is not established. Its 90% actor allocation concerns a Tier-I gap, not demonstrated beyond-time harm.

6 Direct reproduction and artifact availability

The primary reproduction guide links every group, field, command and availability limit. Table 5 gives direct entry points. The protocol, runner, PPO and composer match registration 6262b2b byte for byte. Pinned dependencies and manifests record the numerical environment and source revisions; analysis source is linked at frozen release 74c6ceb.

Checkpoint limitation. Full serialized donor/runner trees were not archived. P/C/D fingerprints cover four-seed *batches*, not downloadable per-seed states, and cannot reconstruct omitted arrays. The frozen runner regenerates donors in memory; the guide distinguishes fresh recomputation from cell resumption. Archived curves permit statistical reanalysis, whereas a new rollout-level reproduction requires recomputation.

Table 5: Direct primary-study artifacts. Frozen scientific-source links point to release 74c6ceb; new CSV/index links point to the presentation release. No donor checkpoint blob is implied by a fingerprint.

item	directly linked artifact
Protocol and statistical rule	Registration, registered analysis implementation
Donor construction and swaps	Onset/fork construction and natural-context composition; actor/critic/Adam field selection
State-integrity checks	Runtime full-tree identity assertions; independent arithmetic/source audit
Every paired training seed	32-row primary CSV; full primary/secondary values and sensitivities
Eight raw groups and manifests	Hashed artifact index, including recorded checks and P/C/D *batch* fingerprints

7 Limitations and conclusion

What the result establishes. Under one declared natural future context, a learner's own pre-impairment actor checkpoint causes a substantial improvement in measured pause-relative recovery. Reciprocal transfer and the specified checks constrain a simple extra-clean-progress or optimizer/world-only explanation. The repair removes most, not all, of this mean cost. It remains a total downstream effect of this field-level intervention; it is not a causal accounting of uniquely stored impairment information.

What it does not establish.

- **Generality:** the strongest confirmation is one Asterix environment implementation, PPO, deterministic reward-sign corruption, small MLPs and one 200-update recovery horizon. It is not a law for RL agents, other architectures, impairments or training budgets.
- **Context independence:** historical pause and matched learner references are different; natural, standardized and reset futures answer different interventions. Positive checks do not establish a context-neutral decomposition.
- **Irreversibility or new-task plasticity:** an AUC cost and a late-window deficit do not prove permanent damage. No experiment measures preserved capacity to learn a new task after impairment.
- **Mechanism or storage:** actor replacement changes future data and learning. Replay time courses, inconclusive label-exposure primaries and policy-KL associations do not identify a unique internal representation or natural storage location of impairment.
- **Collective certainty or deployment:** heterogeneous local test families provide no paper-wide error guarantee or task-prevalence estimate. The fixed evaluation pools and unavailable full donor trees limit reproduction; no automatic checkpoint-selection or deployment rollback policy is validated.

The defensible conclusion is specific: *a past-actor intervention repairs 91.8% of the measured mean native recovery cost in this registered Asterix PPO experiment*, with a small residual and failed matched-state practical equivalence explicitly retained. The findings do **not** establish irreversible damage, preserved plasticity on new tasks, or a unique internal representation of impairment. The framework is useful because it makes these boundaries explicit, not because it turns one successful restoration into a general recovery law.

Broader impact statement

These are controlled small-agent experiments in simulated environments. Potential use in reliable RL recovery requires evaluation of checkpoint selection, new tasks, longer horizons and deployment risks; no safety guarantee follows from this study.

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